

Bayesian Analysis of Colorado Avalanche Tracking Data

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April 24, 2026

Why Bayesian Statistics for Avalanche Data?

Hockey analytics challenge: sparse player-level data. Players take only 30–80 shots per season; raw shooting % is unreliable.

Bayesian hierarchical models solve this via partial pooling: borrow information across players to shrink estimates for low-volume players toward the team mean, while high-volume players deviate meaningfully.

Traditional hockey metrics (goals, assists) are **rare and random** aka poor measures of true player skill.

This project: Separate location-driven xG from genuine finishing skill. Quantify how specific puck actions make a difference on scoring probability.

Relevant Background

xG Models (2012)

First presented at SLOAN Conference. Evolved from shot quality models to probability a shot becomes a goal.

Bayesian Hierarchical xG (EPL)

Scholtes & Karakuş (2023): Player position had minimal impact on xG when shot-related factors were included. Informs our player random effect priors.

Expected Threat / Possession Value (Soccer → Hockey)

Hudl's xT: assigns a value to each zone based on goal probability. Possession Value measures probability of scoring in next 10 seconds from any puck action.

Markov Game Framework

Routley & Schulte (2015): Same pass or carry has meaningfully different values depending on where it occurs on ice, the first model to quantify location × action interaction.

EPV for Hockey (Wisehockey, 2023)

World's first automated real-time EPV for hockey. Measures how likely puck control leads to scoring in next 8 seconds.

About the Data + Data Collection

Source: Provided directly by Colorado Avalanche analytics team (HALO Hackathon) via NHL Edge, infrared sensors in pucks and players' jerseys.

Design: Observational study. 480 AHL games. ~15 million observations across 5 datasets.

Five Datasets: games (metadata) | events (puck actions + coordinates) | stints (line combinations + ice time) | players (identity/attributes) | tracking (raw sensor positioning)

Key features: Every event tagged with spatial coordinates, player identity, game context, and outcome enables deep spatial and sequential analysis beyond traditional play-by-play.

Research Questions

Q1

How does xG (Expected Goals) vary by shot location and player?

xG (Expected Goals): the probability that a given shot results in a goal, based on where it was taken from.

Method: Bayesian hierarchical logistic regression with spatial fixed effects + player random intercept.

Q2

Is the xG relationship stable across time, or does a player's shooting probability drift across a 10-game window?

Method: Binomial mixed model with random slope on 10-game time blocks. Key: σ_{drift} .

Q3

How does each puck action change the probability of generating a dangerous offensive outcome within the next 10 seconds, given spatial and game context?

Method: Bayesian logistic model with action-type fixed effects, spatial controls, contextual features (one-timer, slot pass, etc.), and player random intercept.

Methods

Question 1: xG by Shot Location and Player

Q1 — Data Cleaning

1. Filter events to shots and goals only

Remove passes, zone entries, faceoffs. Focus only on event types that directly produce a shot on goal.

2. Drop missing data

Remove rows with missing x/y coordinates, outcome, or player ID.

3. Create binary outcome

goal = 1 if event type == "goal", else 0. Confirmed ~5% goal rate which is consistent with NHL averages.

4. Standardize coordinates

Scale x and y to mean=0, SD=1. Saving scale parameters for use in Q2 (ensures spatial comparability).

Also reflecting/flip coordinates so all shots attack in the same direction (useful for future plotting)

Final Dataset: 56,787 shot-level observations.

Q1 — Prior Research to Inform Priors

Key paper: Scholtes & Karakuş (2023), Bayesian Hierarchical xG for EPL

"Player position had minimal impact on xG when considering more shot-related factors." → Informs a tight prior on the player random effect SD.

Prior choices:

- Location fixed effects: $\text{Normal}(0, 1.5)$ — zero-centered, allows large location effects.
- Intercept: $\text{Normal}(-2, 1)$ — maps to ~12% baseline shooting %, consistent with observed data.
- Player SD: $\text{Exponential}(2)$ — started at $\text{Exp}(1)$, tightened to $\text{Exp}(2)$ after observing near-zero player variance. Reflects prior skepticism that individual skill drives xG after location is controlled.

Why $\text{Exp}(2)$ instead of $\text{Exp}(1)$? The literature and exploratory analysis both showed very small player differences; $\text{Exp}(2)$ places more mass near zero, which is more accurate for this context.

Q1 — Model Formula & Architecture

Formula: $\text{goal} \sim \text{x_scaled} + \text{y_scaled} + \text{l}(\text{x_scaled}^2) + \text{l}(\text{y_scaled}^2) + (1 \mid \text{player_id})$

Fixed effects: x_scaled, y_scaled (lateral + depth), quadratic terms for curvature in xG surface.

Random effect: (1 | player_id) - partial pooling across players. Each player gets an intercept shifted toward the mean. Low-volume players shrink strongly; high-volume players can deviate.

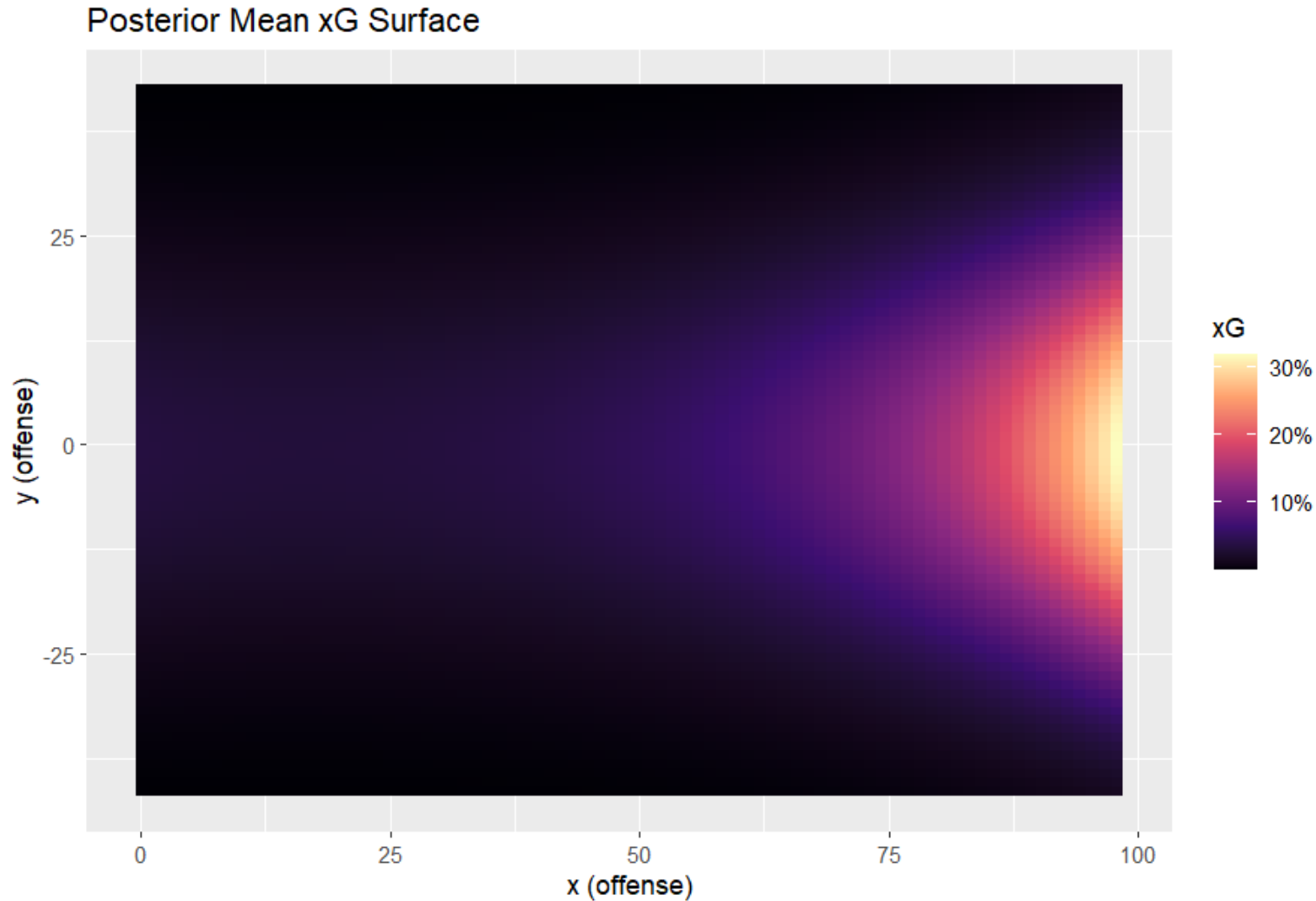
Fitting: brm(), Bernoulli family, 2 chains × 1000 iterations, adapt_delta=0.95, 4 cores.

Diagnostics: All Rhat = 1.00. ESS 600–1200 (lower for player RE is expected near-zero funnel behavior, not a concern). Trace plots: "fuzzy caterpillars." PPC: model goal rate matches observed ~5%.

Results

Question 1: xG by Shot Location and Player

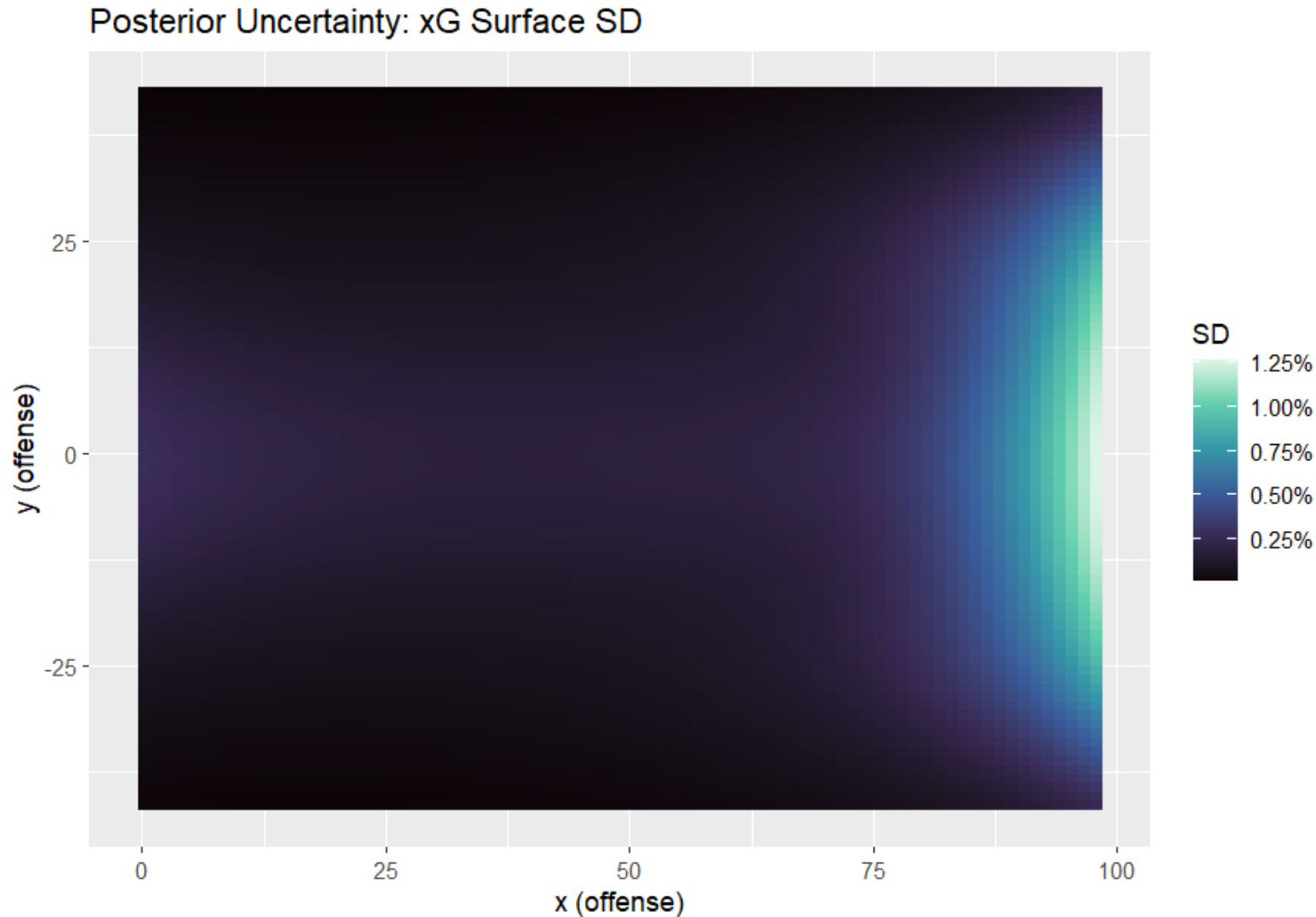
Q1 — Posterior Mean xG Surface



Key finding:

- Hot zone (30%+ xG): directly in front of net, high x, centered on $y=0$
- xG decays sharply with distance and angle
- Corners and perimeter: ~10% xG
- Model corr. with AVS xG: 0.56
- **My mean xG: 0.499**
AVS mean xG: 0.481
(nearly identical)

Q1 — Posterior Uncertainty (xG Surface SD)



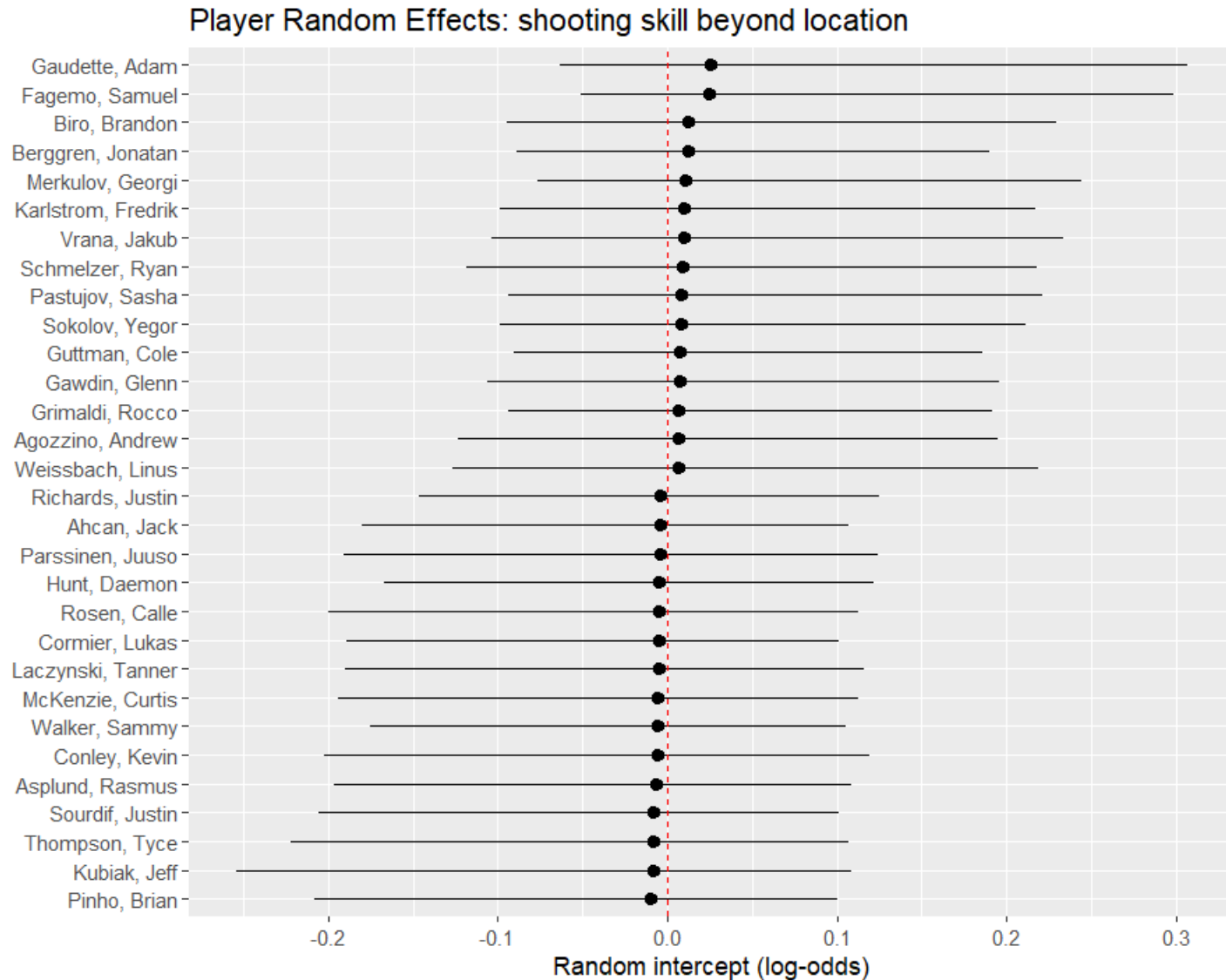
A uniquely Bayesian output:

Uncertainty is highest where xG is highest (near net, SD up to 1.25%)

Near-zero uncertainty on perimeter

The model is most uncertain where outcomes are most variable.

Q1 — Player Random Effects



Three key findings:

1. All CIs wide & cross zero
→ Individual finishing skill indistinguishable from noise at AHL volumes.
2. All estimates within ± 0.05 log-odds of zero
→ Practical magnitude is tiny.
3. Gaudette leads (but wide interval = limited shots)

Conclusion:

Where you shoot from matters far more than who is shooting.

Q1 — How does xG (Expected Goals) vary by shot location and player?

Overarching Conclusion

Shot location dominates xG. Individual finishing skill is negligible at AHL shot volumes. Where you shoot matters far more than who shoots.

Methods

Question 2: Temporal Stability of xG

Q2 — Data Cleaning & Aggregation

1. Start from Q1 cleaned shots data

Same rescaling as Q1 for spatial variables

Join game dates, rank each player's games chronologically.

2. Assign 10-game time blocks

Each shot is tagged to a block. Filter to players with ≥ 2 blocks.

3. Aggregate to player-block level

Each row: player $\times$ block $\rightarrow$ goals, shots, mean x, mean y. Reduces 56,787 shots to $\sim 1,716$ player-block summaries.

4. AR(1) structure

Lagged block variable for temporal autocorrelation.

5. Quality check: Only 73 of 1,716 blocks (4%) had < 5 shots, thus the data is reliable for modeling.

Q2 — Prior Research to Inform Priors

Domain expectation: Shooting behavior is largely stable within a season. Players don't dramatically change which zones they shoot from across 10-game windows. Following Desrosiers (2021), who found AHL shooting profiles translate stably to the NHL

Prior for σ_{drift} : Exponential(2). Rate=2 sets prior mean of 0.5 log-odds
-permissive enough to detect real drift if it exists, but concentrating mass near zero to reflect stability prior belief.

An AR(1) lagged block term is also included to account for natural carry-over between consecutive windows, since blocks 11–20 resemble blocks 1–10 more than they resemble blocks 41–50, without it the model would treat each window as fully independent, which is unrealistic.

Intercept: Normal(-3, 1). Centered at -3 log-odds → baseline block-level goal rate ~4.5%, consistent with observed mean of 1.51 goals per 30.3 shots (~5%). $\log(0.05/0.95) \approx -2.94$ justifies -3 center.

Same location priors as Q1 to maintain spatial comparability across both models.

Q2 — Model Formula & Architecture

goals | trial(shots) ~ mean_x + mean_y + (time_block || player_id)

mean_x + mean_y: Controls for shot location quality per block — connects directly to Q1 xG surface.

(time_block || player_id): The random slope captures how each player's xG-location relationship shifts across time blocks. Slope near zero = stable. Large slope = drifting. || uncorrelated intercept+slope (necessary when both SDs near zero).

Key estimand σ_{drift} : SD of all player-level time slopes. If near zero, the Q1 surface is a reliable seasonal summary.

Fitting: brm(), Binomial family, 4 chains × 3000 iterations, adapt_delta=0.97, max_treedepth=12.

Results

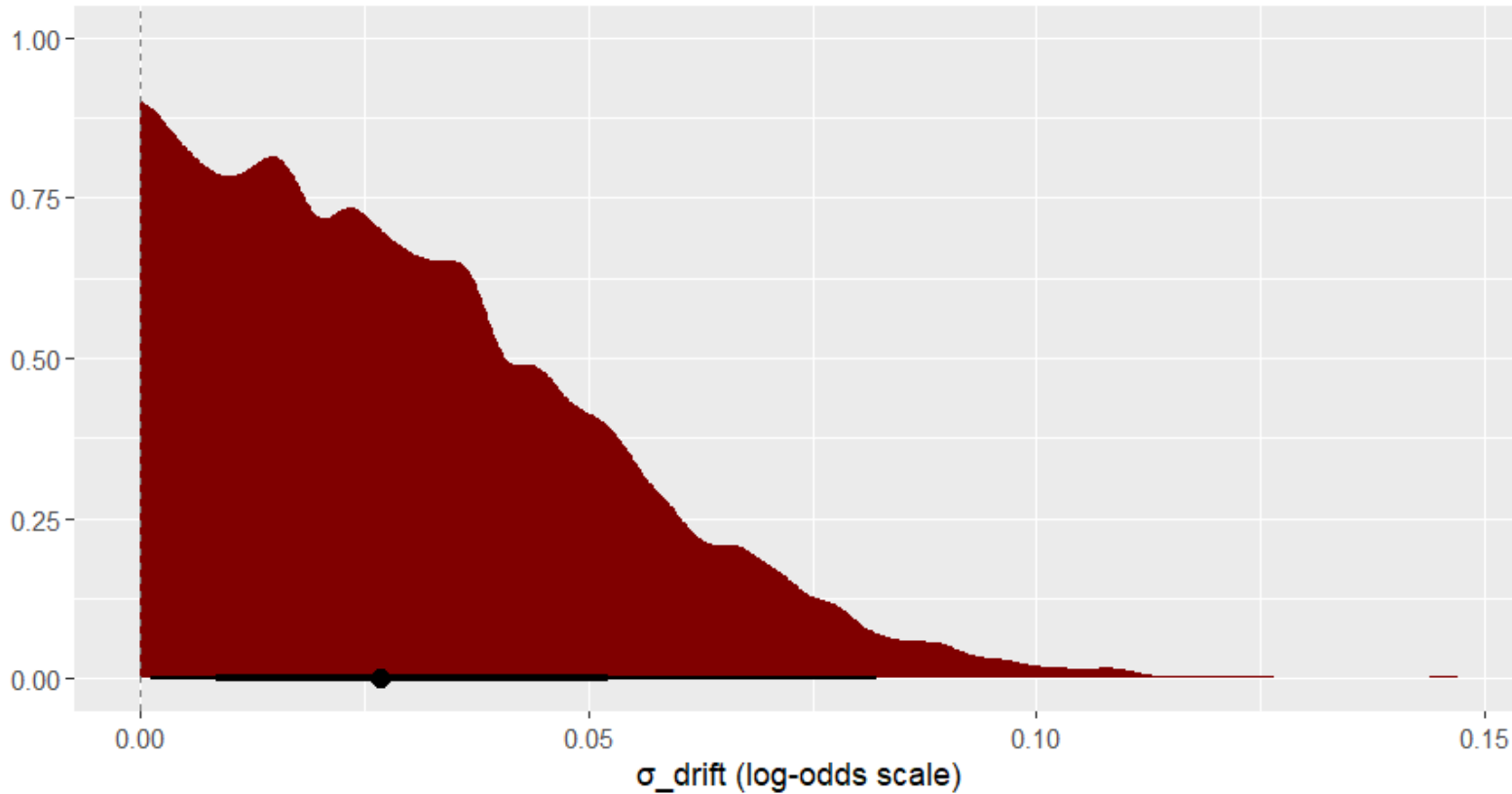
Question 2: Temporal Stability of xG

Q2 — Posterior Distribution of σ_{drift}

Posterior distribution of xG surface drift (σ_{drift})

Near zero = player xG surfaces are stable across the season

Large = players' shooting probability from a given location shifts over time



σ_{drift} estimate:

0.03 (95% CI: 0.00 – 0.07)

Practical meaning:

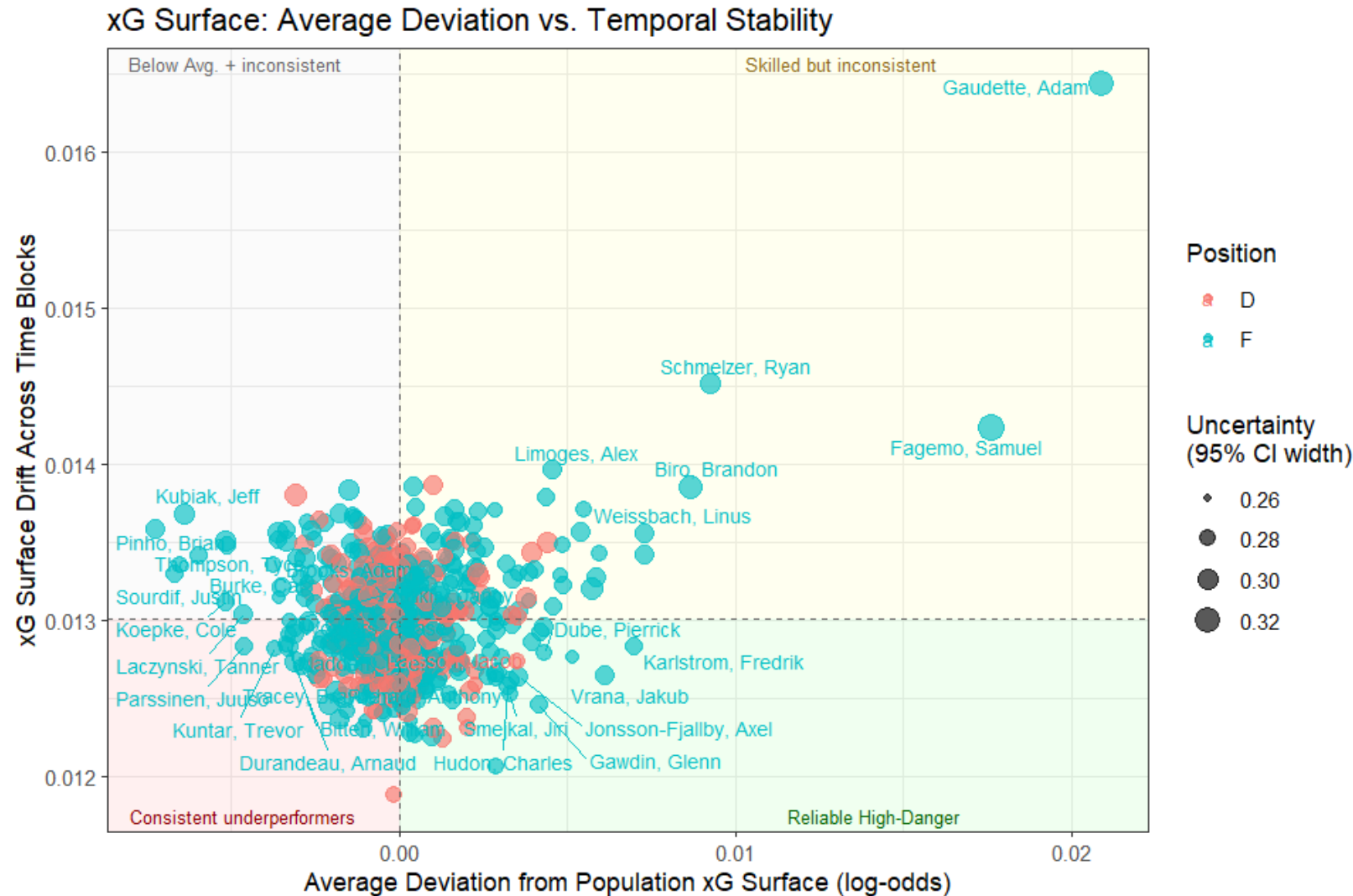
0.10% shift in goal probability per 10-game window

< 1/20th of one goal per 30 shots

Conclusion:

Q1 static xG surface is a trustworthy seasonal summary. Coaches can use it at playoff time.

Q2 — Players Average Deviation vs. Temporal Stability



Four quadrants:

Reliable High-Danger

Gawdin, Vrana: high xG, stable

Skilled but Inconsistent

Gaudette: high xG but drifts

Consistent Underperformers

Durandeau, Kuntar: low & stable

Below Avg + Inconsistent

Kubiak, Pinho: poor & volatile

- stability is universal, not position-specific

Q2 — Posterior Predicted Goals (Next 3 Blocks)-SKIP

Posterior Predicted Goals and Implied xG — Next 3 blocks

Player	Block	Shots	Goals (median)	Goals (lower)	Goals (upper) xG
Fagemo, Samuel	3	76	4	1	9 5.3%
Fagemo, Samuel	4	76	4	1	9 5.3%
Fagemo, Samuel	5	76	4	1	10 5.3%
Gaudette, Adam	4	66	5	1	10 7.6%
Gaudette, Adam	5	66	5	1	11 7.6%
Gaudette, Adam	6	66	5	1	11 7.6%
Gawdin, Glenn	4	47	2	0	6 4.3%
Gawdin, Glenn	5	47	2	0	6 4.3%
Gawdin, Glenn	6	47	2	0	6 4.3%
Vrana, Jakub	3	40	2	0	6 5.0%
Vrana, Jakub	4	40	2	0	6 5.0%
Vrana, Jakub	5	40	2	0	6 5.0%
Smejkal, Jiri	3	22	1	0	4 4.5%
Smejkal, Jiri	4	22	1	0	5 4.5%
Smejkal, Jiri	5	22	1	0	4 4.5%

Key insight:

Shot volume drives forecast reliability more than skill.

Fagemo (76 shots): range 1–10

Smejkal (22 shots): range 0–5

Flat forecasts across blocks validate

$\sigma_{\text{drift}} \approx 0$

— no trend up or down.

Q2 —Is the xG relationship stable across time, or does a player's shooting probability drift across a 10-game window?

Overarching Conclusion

The xG surface is temporally stable ($\sigma_{\text{drift}}=0.03$). Coaches can rely on single-season xG estimates at playoff time. There is no meaningful drift across 10-game windows.

Methods

Question 3: Puck Action → Danger Within 10 Seconds

Q3 — Data Cleaning & Feature Engineering

1. Define dangerous outcome

For each event, look ahead ≤ 10 seconds for a shot/goal by the same team. Binary flag = 1 if found.

2. Convert time to absolute seconds

Enables cross-period time gaps. Critical for 10-second window calculation.

3. Remove non-puck actions

Drop faceoffs, whistles, period endings, and actions not chosen by players.

4. Feature extraction from flag/detail columns

Binary indicators: one-timer, seam, slot pass, rush pass, zone entry, slot shot, is_successful.

5. Collapse rare event types → "Other" | Pass = reference category (26.5% danger rate, near middle)

Validated: reception 33%, carry 31%, block 13% ✓ , makes intuitive hockey sense.

6. Subsample 20% (~200K obs) for computational tractability.

Preserves 21.95 outcome rate.

Q3 — Prior Research to Inform Priors

Intercept: Normal(-1, 1)

→ $\log(0.219/0.781) = -1.27 \approx -1$. Calibrated directly to 21.9% observed danger rate.

General action effects: Normal(0, 1.5)

→ Deliberately wider than Q1/Q2. Decroos et al. (2019) found action type explains more variance than location alone in possession-value frameworks.

Slot/rush features: Normal(0, 2)

→ Routley & Schulte (2015): slot and rush situations generate disproportionately high possession value. Prior permits larger coefficients.

Player Random Effect SD: Exponential(2)

→ Simpson et al. (2017) principled complexity penalty. Skepticism that large player differences remain after controlling for action, location, and context.

Relevant Definitions

Carry: a player skating with the puck rather than passing it, actively driving toward the net.

Loose Puck Recovery (LPR): regaining possession of a puck that neither team currently controls, like after a blocked shot or collision.

One-Timer: a player receives a pass and shoots in one motion without stopping the puck first (is_timer)

Slot: the high-danger rectangular area directly in front of the net, roughly between the faceoff circles.

Rush Pass: a pass made during a rush situation, where the offense outnumbered or matches the number of defenders (is_rush_pass)

Zone Entry: the moment a team crosses the opponent's blue line into the offensive zone with possession. (is_zone_entry)

Seam Pass: a pass threaded through a narrow gap between defenders into a dangerous central area. (is_seam)

Q3 — Model Formula & Architecture

Danger ~ action_type + x_scaled + y_scaled + is_onetimer + is_successful + is_slot_pass + is_rush_pass + is_zone_entry_is_slot_shot + is_seam + period + (1 | player_id)

Key variable: action_type (pass = reference). Each coefficient = change in log-odds of danger vs. a pass.

Contextual enrichment: 7 binary features capturing one-timers, slot actions, rush, zone entry, action success, seam passes.

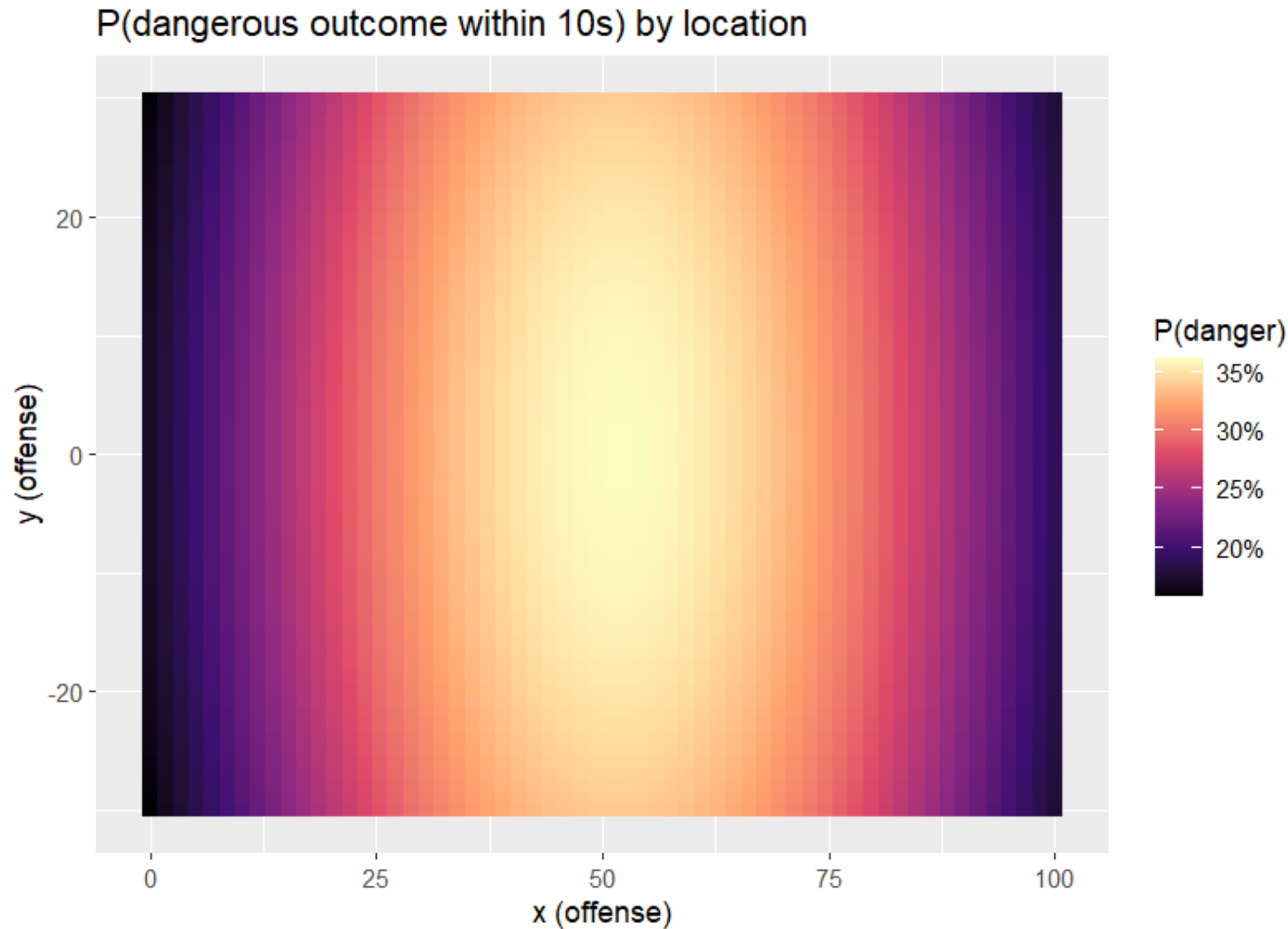
Period control: Game state varies P1–P3; teams play differently as time runs down.

Fitting: 2 chains × 1500 iter, Bernoulli family, adapt_delta=0.95, max_treedepth=12.

Results

Question 3: Puck Actions and Danger Probability

Q3 — P(Danger) by Location



Key contrast with Q1:

Q1 surface peaks right in front of net (high x).

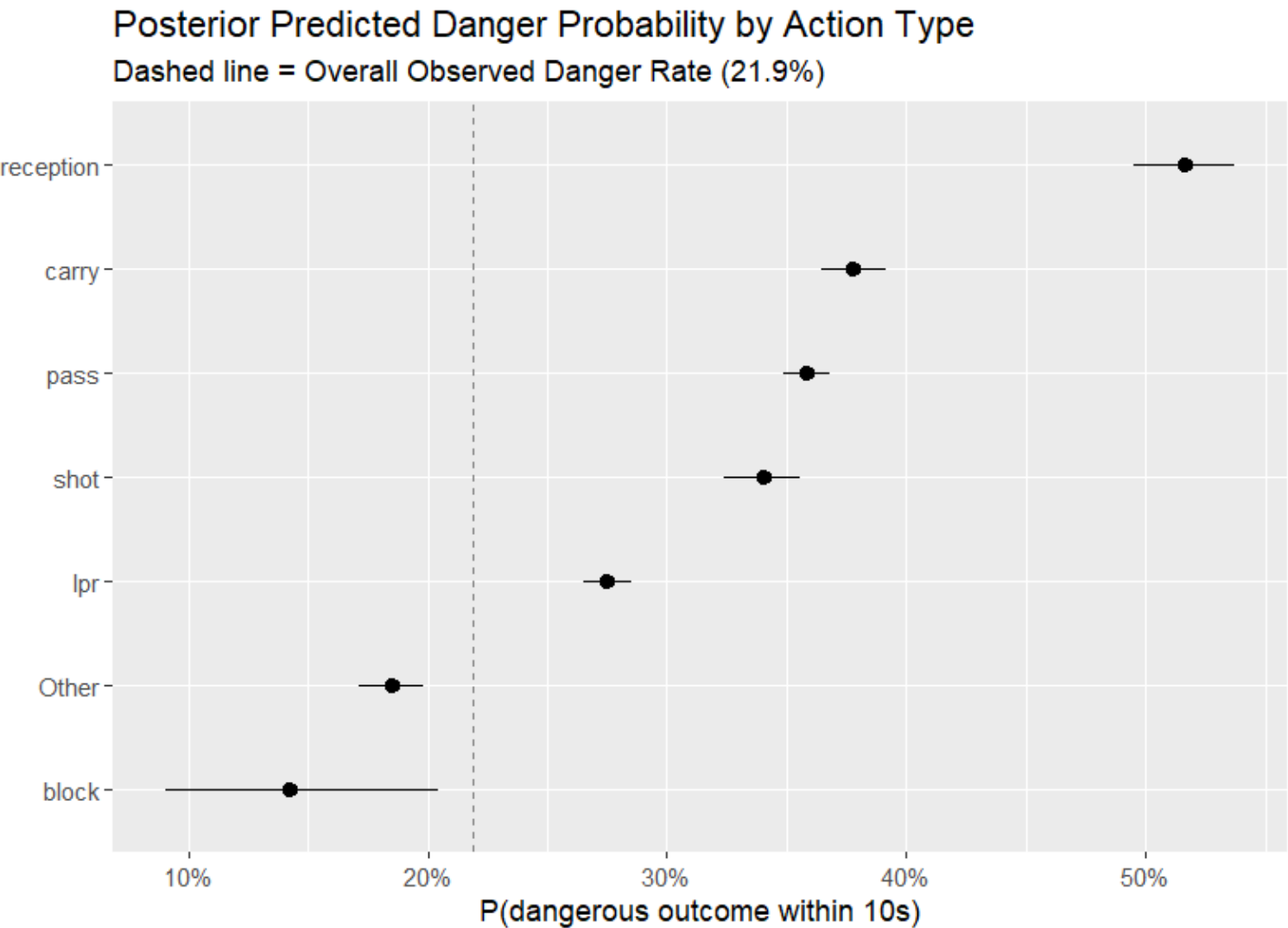
Q3 surface peaks in the MIDDLE of the offensive zone (x≈50).

Why? Middle zone = transition point. Actions there most frequently precede shots within 10s.

Implication:

Where you ACT matters less than HOW you act.

Q3 — Posterior Danger Probability by Action Type



Reception: 52% danger
Just received pass = ready to shoot

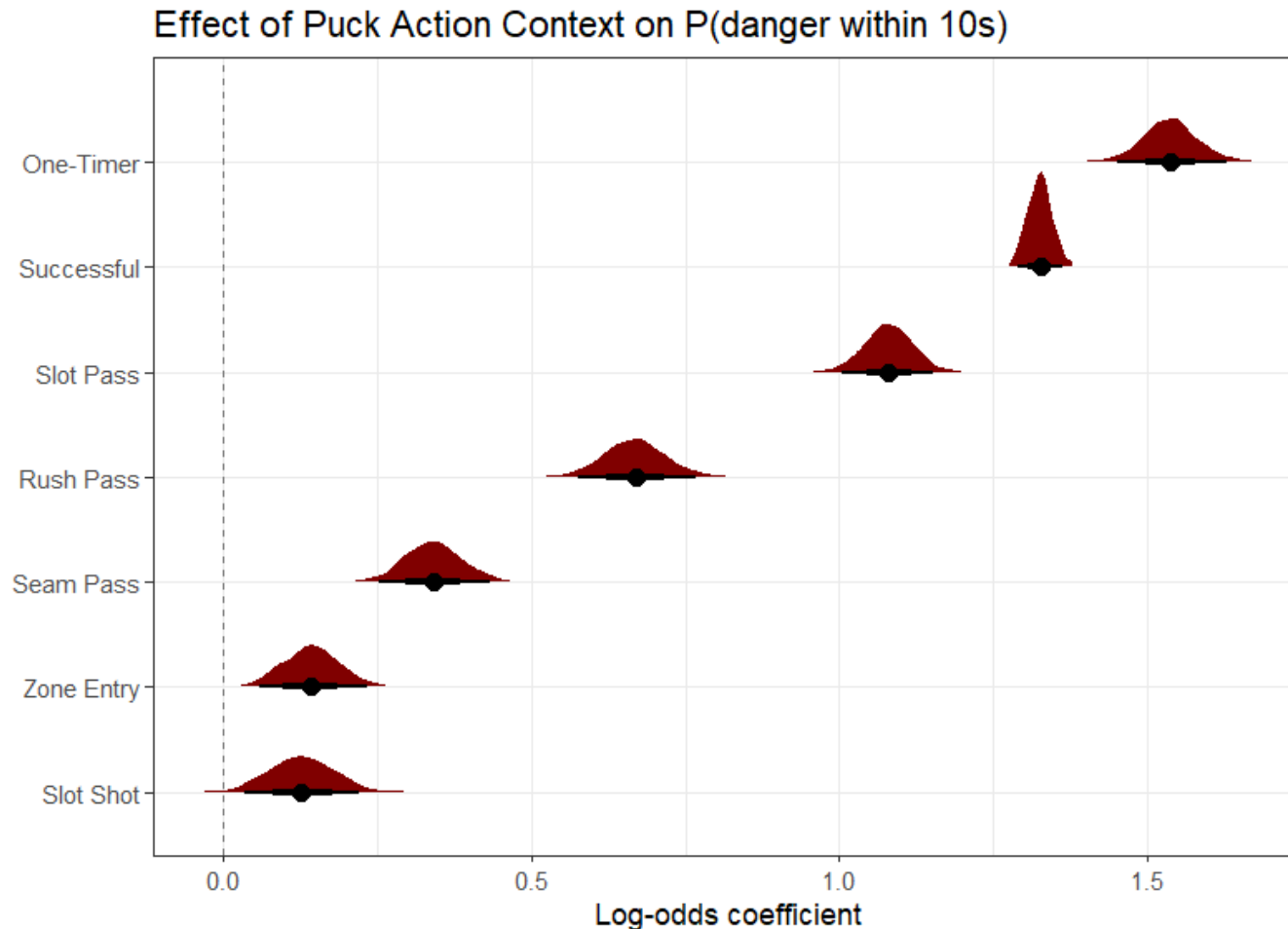
Carry: 38%
Driving toward the net

Pass (reference): 36%
Above the overall rate.
-offensive zone passes set up shots

Block: ~14%
Defensive, danger drops immediately

Overall rate: 21.9%

Q3 — Effect of Puck Action Context



One-timer: +1.54 log-odds

+15–20% danger. Dominant predictor.

Action success: +1.33

Nearly as predictive as a one-timer.

Slot pass: +1.08

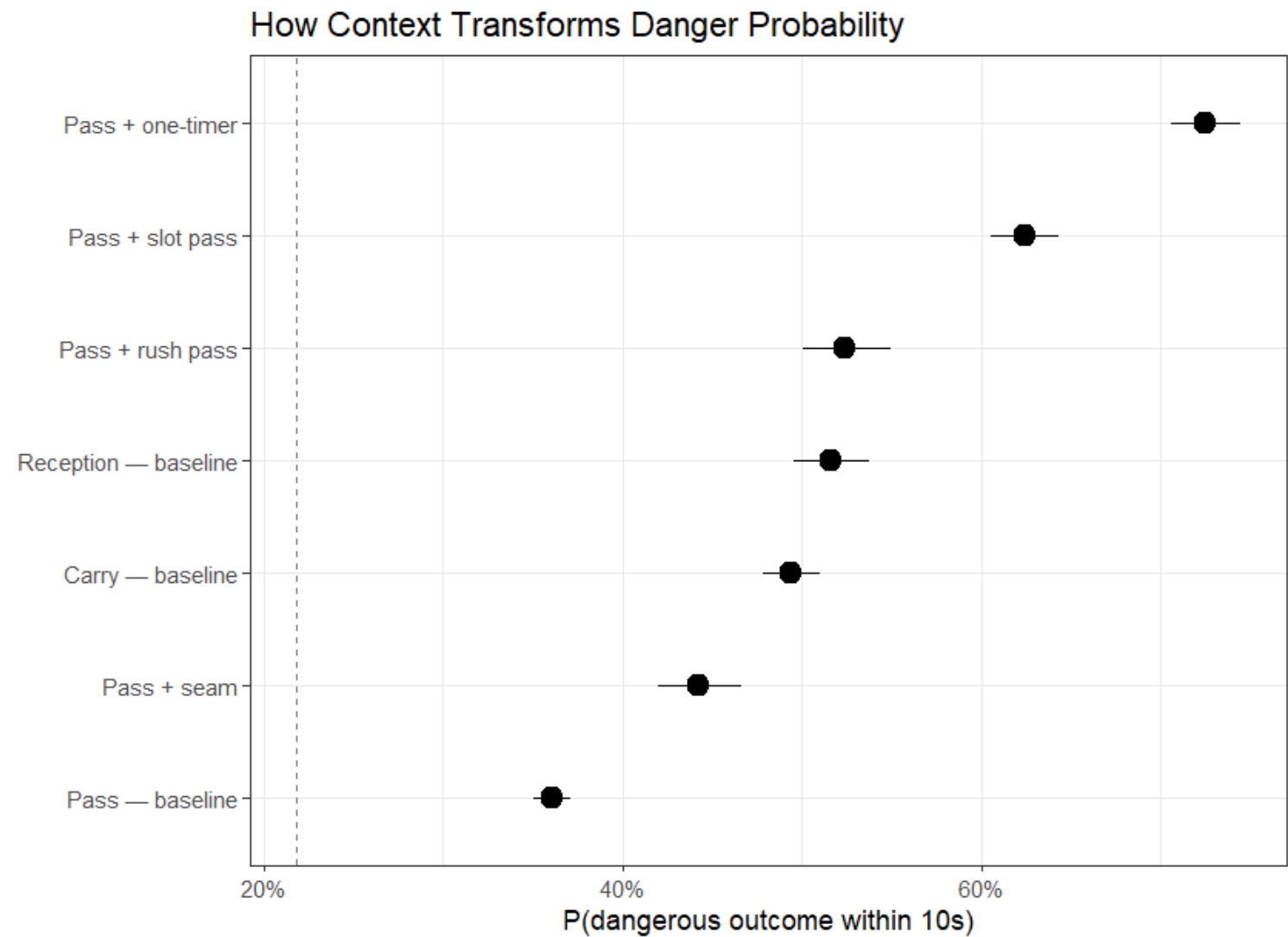
Directly sets up high-quality shot attempts.

Rush pass: +0.67

Transition advantage captured.

All features increase danger.

Q3 — How Context Transforms Danger Probability



Key finding:

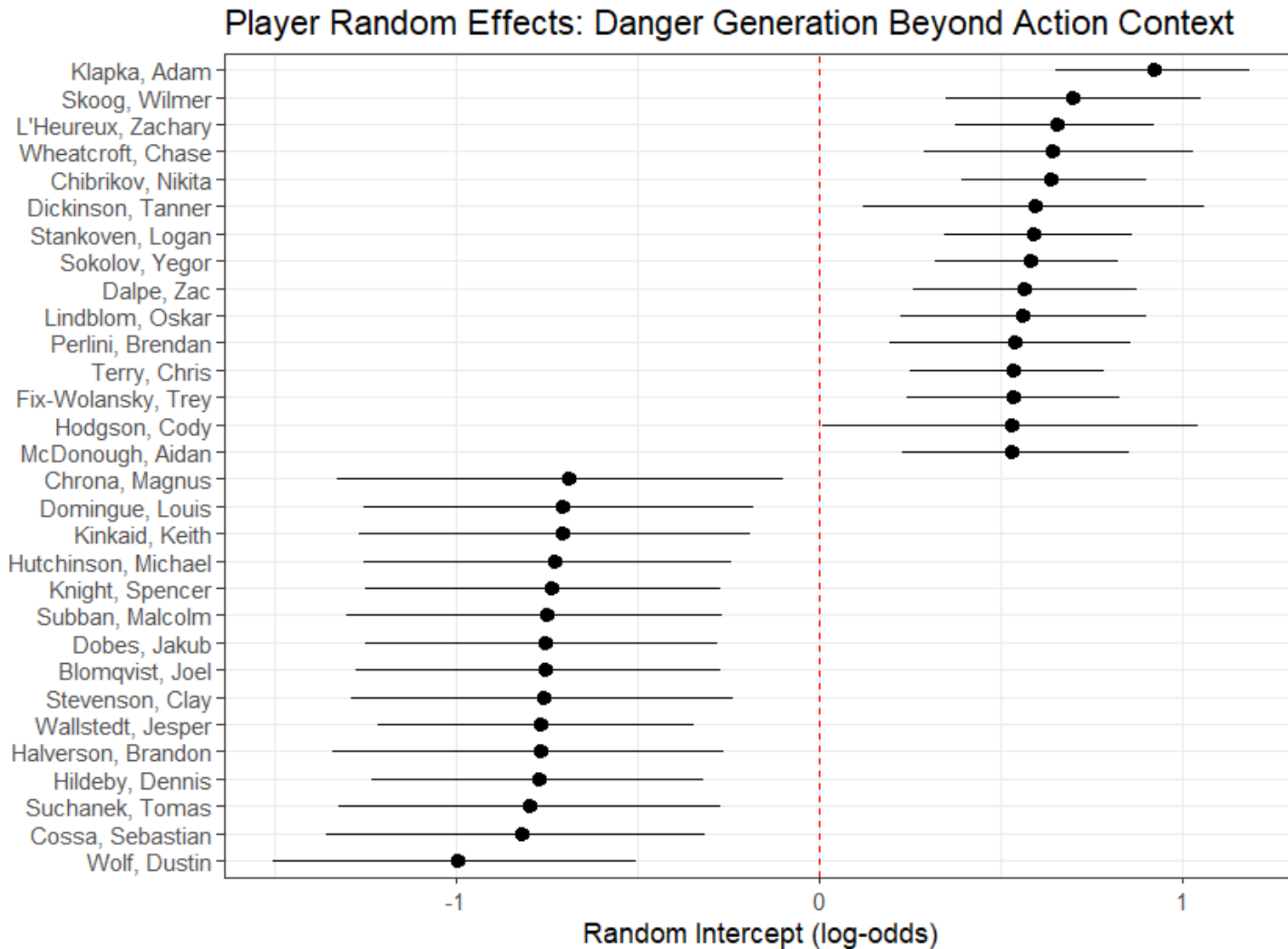
Pass — baseline: 33%

Pass + one-timer: 65%

+32 percentage point increase from one context feature!

The same action can have radically different danger implications depending on context.

Q3 — Player Random Effects: Danger Generation



Critical contrast with Q1:

Q1 player SD: ~0.05 (negligible)

Q3 player SD: ~0.37 (substantial)

Top generators:

Klapka, Skoog, L'Heureux, Wheatcroft

Top suppressors:

Wolf, Cossa, Suchanek

(note: Wolf is a goalie)

Skill type matters:

Finishing skill $\neq$ Danger-creation skill

CONCLUSIONS

Concluding Thoughts

Q1

Shot location dominates xG. Individual finishing skill is negligible at AHL shot volumes. Where you shoot matters far more than who shoots.

Q2

The xG surface is temporally stable ($\sigma_{\text{drift}}=0.03$). Coaches can rely on single-season xG estimates at playoff time. There is no meaningful drift across 10-game windows.

Q3

Context transforms danger probability more than action type.
A pass → one-timer generates 65% danger vs. 33% for a baseline pass. A 32% gap from one feature.

Key
contrast

Finishing skill (Q1) is invisible once location is controlled. Danger-creation skill (Q3) is substantial and real (player SD 0.37) is seen as a new, actionable metric.

Insights-Based Recommendations

For Coaches

Design systems that generate slot passes and one-timer opportunities. Even a moderate increase in these high-context actions can dramatically shift danger probability (+32% per one-timer context).

For Player Evaluation

Use danger-creation skill (Q3 random effect) alongside location-adjusted xG (Q1) for a more complete picture. Finishing skill $\neq$ danger-creation skill. They measure different abilities.

For Roster Construction

Distinguish players who score because they find the right spots (location-driven) from those who create danger regardless of context. The Q3 random effect quantifies this second ability.

For Analytics

Single-season location-adjusted xG is reliable for projection (Q2 validation). No need for mid-season recalibration. Use posterior credible intervals when evaluating low-volume players.

Limitations & Caveats

1. Independence assumption violated

Shots within a game are not independent. A full Markov game model would account for sequential dependencies. This is a key limitation in Q1 and Q3.

2. Subsampling in Q3

20% subsample preserves outcome rates but may under-represent rare action combinations. Full dataset would yield tighter posteriors.

4. Missing interaction terms

No $x \times y$ interaction in Q1. No action type $\times$ location interactions in Q3. These could improve the spatial models.

Future Recommendations

1. Full Markov game framework

Model the full sequence of puck actions as a Markov decision process, removing the independence assumption and capturing how plays chain together.

2. Extend to defensive metrics

Q3 currently values offensive actions. A parallel model for defensive disruptions (blocks, interceptions) would give a complete possession-value framework.

3. Real-time EPV

Use the Q3 model as a foundation for real-time expected possession value. Updating after each action during live games for bench-side decision support.

THANK YOU!

*Sydney Stanton | Department of Statistics & Data Science | University of Colorado Boulder
Data provided by Colorado Avalanche Analytics via HALO Hackathon*